Requirement Analysis

Technology Stack (Architecture & Stack)

Field	Details
Team Lead	Konkimalla Bhagyasri Lasya
Team Members	Bhavitha Medakayala, Yogiswari Sanapala, Divya kauluri, Nagaram praveenya
Domain	India's Agriculture Crop Production Analysis
Tools Used	Tableau
Dataset	Historical Indian Crop Production Data, 1997–2021
Published Dashboard	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

Table 1 — Components & Technologies

Component	Technology Used	Purpose
Data Source	CSV / Excel (.xlsx)	Raw crop production data (1997–2021, 345,407 records)
Data Cleaning	Python 3.12 (Pandas library)	Column renaming, null handling, year extraction, Yield calculation
Data Visualization	Tableau Desktop / Tableau Public 2026.2	Building all chart types, dashboards, and the Story
Geographic Mapping	Tableau built-in Mapbox integration	State-wise filled map of India
Dashboard Publishing	Tableau Public (free cloud hosting)	Browser-accessible dashboard sharing via public URL
Version Control	GitHub	Project repository for all deliverables and documentation
Documentation	Microsoft Word (.docx)	Project deliverables, final report, templates
Project Management	Personal task tracker / SkillWallet platform	Epic/story tracking and mentor evaluation

Table 2 — Application Characteristics

Characteristic	Description
Open Source Tools	Python (Pandas) is open-source; Tableau Public is free for public publishing
Security	No PII data; all data is publicly available government agricultural statistics

Scalability	New years of data can be appended to the CSV and the Tableau workbook refreshed without redesign
Availability	Tableau Public guarantees high availability; no server maintenance required
Performance	345K rows render within 3 seconds across all filters on a standard laptop (i5, 8GB RAM)
Compatibility	Works on Chrome, Edge, Safari; responsive on desktop and Tableau Public mobile view